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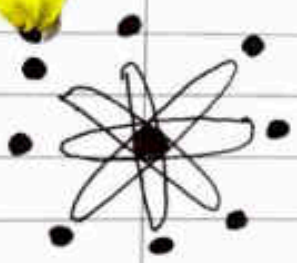




CHRIST (Deemed to be University)
Department of Statistics & Data Science
STA101-1 Descriptive Statistics
CIA 1 Component 1

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Course: Computer Science & Statistics
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Q Explain your answer by discussing the important aspects of statistics in atleast three modern fields such as healthcare, sports, business, climate science, agriculture, banking or education.

A Statistics is used as a means of developing sound techniques for the handling and analysis of numerical data to draw inferences from them in the modern context. It finds wide applications in nearly all modern fields. It has become an indispensible tool for humans.

It gathers massive importance in fields including but not limited to:

I) Business

In business contexts statistics provides a tool for production control. A business more or less depends on the accuracy and precision of forcasting, that is done through statistics. Businessmen utilise statistics to study the needs and desires of the

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consumers. If the businessman is incorrect analysis, they may face financial disaster. For instance, say a businessman wants to manufac ready made garments, he needs to analyse the number of garments to be manufactured thro statistics to avoid manufacturing dead (ext Stock or leaving money on the table because of a lack of stock.

II) Healthcare

In medical sciences, statistics helps greatly in the collection, presentation and an observations of the cause and incidence of va psychological and physiological diseases.

It is also utilised to analyse the resu obtained through the various drugs and medicines used in treatment. The efficacy of a manufactured drug or medicine or injection is also found through statistics.

Statistics shines brightly in the field of mental health. It is used to find

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the reliability and validity of psychological tests. A new subject named 'Psychometry' was born due to statistics' immense influence on psychology.

Statistics has also led to biological advancements that have improved healthcare.

III) Sports

Statistics has three major applications in sports. It can be actively utilised to improve individual progress through analysis of various datapoints at intervals. This data can be used to find trends in sport performance, markers of improvement and areas in need of attention.

Another application is to build game plans by analysing the opponents and own strategies and history. This can be used to both predic the upcoming moves of an opponent or to work around the strengths of an opponent in advance.

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Lastly, the sports industry is heavily dependent on prediction markets in the modern era. Statistics can be utilised make the most precise possible predictions ~~to~~ results of any sport.

IV) Agriculture

In modern and commercial farming, statistic is used heavily to estimate various factors o agriculture.

Statistics is used to plan the number of seeds to be sown. Statistics is also utilised to map out irrigation and rainfall factors.

Statistics is now being used extensively in large-scale operations of agriculture to both i) predict produce
ii) estimate needs of crops and animals.

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Q Explain why statistics can sometimes produce misleading conclusions.

A Statistical conclusions can be misleading when data is collected, analysed or presented incorrectly.

One of the major reasons for misleading statistical conclusions is biased or incomplete data. Data collected from one subset of the population cannot ~~se~~produce ~~that~~ conclusions that would hold true for the entire populations. For instance, if a statistical analysis is being conducted about internet usage in India through an online survey, it will disbar those without the internet to participate and ergo, skew the numbers to be inflated to show more internet usage.

Similarly, small sample sizes are ineffective in representing a large group of people. In small sample sizes, individual biases or

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exceptions may vastly skew the results of the analysis.

In statistics, averages are often misused. Averages hide important variation in data. For example, if one employee is being paid 5 lakh while 5 others are paid 20,000, the average income is 1 lakh for each employee, completely disregarding the 25x pay gap. These averages can give false impressions.

Some graphs and charts will omit certain information to present data in a manner more favourable to the presenter. In a scenario where sales have jumped from 120 to 140, if put bluntly it will show a very small increase, however if we truncate the vertical axis to start from 100, it suddenly becomes a 2x jump on the chart/graph.

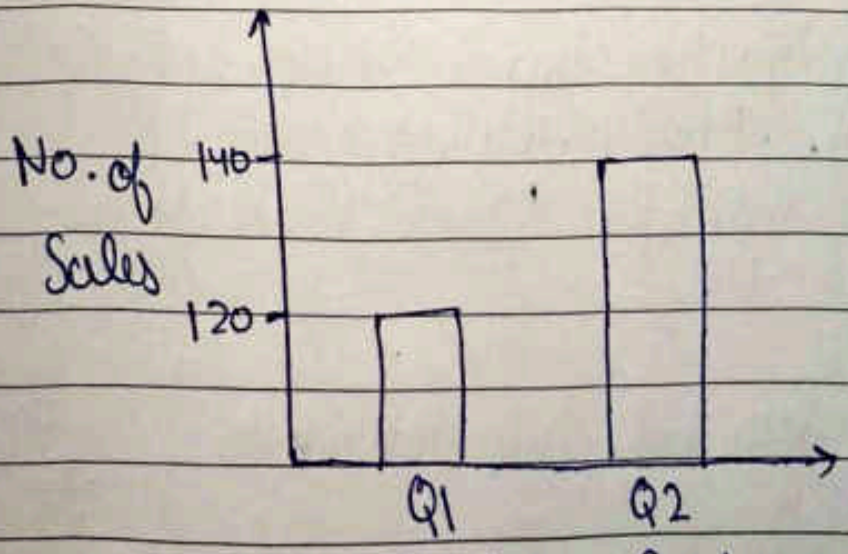
Statistics also suffers from correlation without causation. Many times two statistical data will

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appear congruent. This does not mean that those are related in any manner. For example, ice cream sales and deaths by drowning both go up in summer but have no relation whatsoever.

Sometimes statistics is intentionally manipulated and presented to support certain opinions or agendas. An advertisement could claim "90% effectiveness" without referencing the conditions of testing or the small sample size.

Example of misleading graph:



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Q Discuss any four limitation of statistics with suitable real-life examples.

A Statistics is an indispensible tool for analysing and interpreting data. However, It too has limitations. Some of which are discussed as follows:

I] Statistics is not suited to the study of qualitative phenomenon

Since it deals with numerical data primarily, it cannot directly study qualities such as honesty, intelligence, beauty, culture or poverty because they cannot be measured in numbers.

A person's honesty cannot be expressed numerically. However, intelligence can be studied indirectly through marks obtained in an IQ test.

II] Statistics does not study individuals

Statistics deals with aggregates or groups

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rather than individual cases. A single observation has little meaning unless it is compared with other observations.

Example: Knowing that one student scored 85 marks tells us very little. However, when compared to the marks of an entire class, we are able to paint a better picture.

III) Statistical laws are not exact

The conclusions drawn from statistics are probabilistic and not guaranteed. There is always an element of chance.

Example: A weather report saying that its a 70% chance of rain tomorrow, it does not mean that it 'will' rain but that rain is likely.

IV) Statistics is liable to be misused

Statistical data can be manipulated or interpreted in incorrect manner drawing invalid conclusions from the statistical analysis.

Example: A company may claim that 90% of people prefer their product after surveying just 10 people.

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Q Explain how statistics has evolved from ancient times to the modern digital era by mentioning atleast five important milestones

A Statistics has evolved from simple record-keeping to a modern scientic discipline used in almost every field over thousands of years.

I] Ancient Civilisation and State Administration

The word statistics is derived from the Latin word 'Status', Italian 'statista' or the German 'statistik'. All of these words mean political state.

In these times, the government collected information on population and wealth to administer the state, collect taxes, maintain security.

II) Development of Statistics in Ancient India

According to Arthashastra, during the reign of Chandragupta Maurya, India had an efficient

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system of collecting official statistics such as records of births, deaths, and administ activities were maintained.

Later, during the reign of the Mughals, records of land revenue and agricultural statistics were also maintained.

III) Origin of Vital Statistics in the Seventeenth Century

During the 17th Century, the scientific study of births and deaths began. It was known as Vital Statistics.

John Graunt, the father of Vital Statistics started this by analysing birth and death records in London. ~~IT IGT~~

In 1698, the first life insurance institution was established in London based on the developments of Vital Statistics.

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IV] Development of Probability Theory

In the mid-seventeenth century, mathematicians such as Blaise Pascal and Pierre de Fermat solved the famous 'Problem of Points', laying the foundation of Probability.

This led to the development of the theory of probability.

V] Rise of Modern Statistics in the Twentieth Century

The works of several mathematicians progressed statistics further and made it a major scientific discipline

- Francis Galton pioneered regression analysis
- Karl Pearson developed correlation analysis and the chi-square test
- William Sealy Gosset introduced Student's t distribution
- Ronal Fisher (Father of Statistics), developed analysis of variance and design of experiments.

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VI] Statistics in the Modern Digital Era

Statistics is now the foundation of data science, artificial intelligence, machine learning, business analytics, and big data.

Conclusion

Statistics has evolved since the beginning of civilisation to today. Now, it has become an indispensible tool for the modern era.

Thank You!